

Executive Summary

This project centers around remedying the rising churn rates at Telkom Inc. by shifting the company towards data-driven, proactive retention strategies. Currently, Telkom is facing losses in recurring revenue because customer dissatisfaction is not recognized until it's too late. With the goal of maximizing customer lifetime value, this approach uses predictive modeling to identify high-risk accounts before they churn as well as descriptive analytics to understand why customers leave and develop strategies to address these factors. By providing an end-to-end analytics solution, this project delivers clear benefits to key stakeholders. The CFO gains justification to reallocate investment from customer acquisition to retention that's backed by data, customer service and marketing teams gain actionable tools for proactive engagement, and high-risk customers benefit from an improved service experience and tailored incentives.

Our analysis uses thousands of customer records extracted from current and historical Telkom databases, with each record detailing customer demographics, tenure, service usage, billing information, and churn status. Beyond the necessary transformations required by certain algorithms, the current state of the data in Telkom's databases needed minimal manipulation. Descriptive logistic regression and decision tree models developed on all available data were used to learn more about what determines churn behavior. These models revealed the crucial insight that demographics don't drive churn; customer and service experience do. They indicated that customers with month-to-month plans, electronic check payments, high charges, and subscriptions to internet service were more likely to churn, whereas those with longer tenure, long-term contracts, and engagement with tech support were less likely to churn.

To develop the predictive model that identifies high-risk customers, we refined and tested four different model types: decision trees, random forests, neural networks, and support vector machines (SVMs), each having their respective strengths. The threshold to be determined high-risk is a 25% probability of churn, a level that optimizes sensitivity (correctly identifying churn cases) while not compromising overall accuracy. The models prioritize sensitivity, as the cost of losing recurring customer revenue outweighs that of retention (Kumar, 2022). The decision tree is the predictive model deliverable for this project, as it delivered the highest sensitivity at 79.09% while still maintaining a 75.84% overall accuracy. The balance of sensitivity and interpretability makes it an ideal tool for Telkom's operational needs.

Deploying the predictive model involves scheduling it to run regularly (ex. weekly) over Telkom's customer databases. Accounts that meet the 25% threshold will be tagged as high-risk on their customer profile. On Telkom's CRM software, these tags will display visual flags and tailored intervention scripts for customer service representatives based on the Retention Playbook, which uses information from the descriptive models to map churn drivers to approved promotions and scripts specifically designed to combat that driver. On marketing platforms, these flags will trigger the customer's enrollment in targeted retention campaigns, such as offering "Sign & Save" promotions to customers who switch to a long-term contract. Outside of customer engagement, the IT and network teams must focus on improving service reliability to protect services bundled with internet.

To ensure that Telkom can continue to benefit from this solution over time, the analytics team will lead the model lifecycle management process. This will involve directly updating the models as Telkom's service offerings evolve and re-evaluating all model types to determine if there is a better fit as the data changes. The marketing and customer service teams will also collaborate to ensure that strategies align with shifts in the market. Business benefit will be measured through metrics such as overall churn rate reduction, retained recurring revenue (i.e. from preventing a churn), and the net return on retention campaigns (saved lifetime value versus cost of promotions), which represent financial ROI and operational impact. Results of A/B testing of the new data-driven strategies and information from exit surveys will also be used to determine the strategies' contribution to retention and then incorporated into future model versions to ensure that strategies align with changing customer needs. By deploying this end-to-end solution and intentionally monitoring it, Telkom can continue to refine retention tactics to help maximize customer tenure and minimize churn.

1. Project Purpose

This project is focused on transforming Telkom Inc. from reactive to proactive and data driven in order to prevent customer churn. With customer acquisition costs (marketing, promotions, etc.) exceeding the cost of retention by five to seven times (Kumar, 2022), it is critical to limit customer churn as much as possible in order to avoid the high acquisition costs. To achieve this, we are using an analytical approach in which we develop both descriptive and predictive models to identify these high-risk customers while also developing specific retention

strategies. The descriptive model highlights feature importance, enabling us to identify which variables are most critical in determining whether a customer churns. This allows us to develop tailored retention strategies based on the factors revealed in this analysis. The predictive model takes customer information and allows Telcom Inc. to identify customers at a high risk of churning, allowing the company to proactively intervene using the tailored retention strategies.

2. Data

The dataset used in this project consists of 7043 customer records, extracted from Telcom Inc.'s databases of current and historical customer information. Each record details customer demographics, their tenure with Telcom Inc., service usage, billing, payment, and churn status. Including a unique identifier for each customer, there are 21 variables. A table providing a brief description of each of these variables can be found in the appendix (Table A1). The target variable for our modeling is churn, which indicates whether a customer has cancelled their service with Telcom Inc.

2.1 Cleaning and Transformation

The data is exceptionally clean and perfectly structured for model development. Each categorical entry aligns with preset definitions, and numeric variables contain no extreme outliers that would suggest data entry errors. The only missing entries occur in the total lifetime charges field, but investigation confirmed that these missing values occur only for new customers with a tenure of zero who have not yet completed a monthly billing cycle. These missing values were replaced with zeroes to allow the models to function properly. Of the 21 variables, the unique customer ID was the only one removed as it is not useful in our analysis.

Although the data is formatted well, the data must be transformed for certain models such as neural networks and SVMs. Numeric data was standardized such that each column has a mean of zero and a standard deviation of one. This ensures that variables with larger values (ex. total charges) don't skew the model. For categorical variables, each unique value within a column becomes its own individual binary column, filled in with a 1 if that characteristic applies to that customer record and a 0 if not.

3. Descriptive Models

3.1 Model Development

The descriptive models utilize all available data in order to determine the factors that are historically most relevant in determining whether a customer churns. The logistic regression model was developed in R by starting with a baseline model that includes every variable. This full model was very large, so it was refined using backward elimination as a variable selection technique. This refined model excludes statistically irrelevant and redundant variables. All code is available for Telcom to replicate this analysis.

The decision tree model was developed in a similar way with a baseline model followed by refinement. The complete dataset was fed into R to grow the decision tree to its maximum possible size. R output includes the classification error that would occur if tree growth was stopped at a given split point. The full tree was pruned such that we are left with the smallest tree possible that has an error less than the sum of the error of minimum error tree and its standard error. This reduced the full tree from 1345 splits to 8 splits, while maintaining accuracy.

3.2 Feature Importance

The logistic regression and decision-tree models reveal the most important factors in determining whether a customer churns. In the first project checkpoint, the logistic regression model was determined to not fit the data well. However, after seeing that the logistic regression model highlights almost all of the same factors as the decision tree, we decided to also include factors only supported by the logistic regression model (as well as factors only supported by the decision tree). Important factors were determined based on the chi-squared test significance levels for the logistic regression model and feature importance rankings for the decision tree. Below is a table that highlights the tendencies of customers who churn and of customers who don't.

No Churn	Churn
- Longer tenure - One/two-year contracts - Using tech support	- Internet service (especially fiber optic) - Higher total charges - Multiple Lines - Payment via electronic check

Table 3.2.1. Customer tendencies by churn type.

Table 3.2.1 includes two tendencies that may seem contradictory but are easily explained. The models tell us that longer tenure decreases churn risk, but higher total charges increase it. Logically, these numbers should increase together as a longer tenure allows for a greater accumulation of charges. But it can be thought of as a tie breaker. While higher tenure customers are less likely to churn, for people with the same tenure, those with higher total charges may be more likely to churn.

4. Predictive Models

4.1 Data Partitioning and Testing Strategy

To ensure the predictive models can accurately identify high-risk customers, the models must be tested on “outside” data. The dataset was randomly split into training and testing groups, with 80% of the data being used to build and tune the models, and the remaining 20% used as a testing set. Each model type uses the same train and test split.

Each of the predictive models produces a probability of customer churn given a data record. Typically, a probability of 0.5 is used as a threshold for classification, which in this case would mean above 0.5 churns and below 0.5 doesn't. This project evaluates each model at multiple classification thresholds, ranging from 0.2 to 0.5, to find an optimal balance between accuracy and sensitivity. Recall that sensitivity measures a model's ability to correctly identify positive cases, or in this case customers who are more likely to churn – the priority of the predictive model. Although this may involve a tradeoff in specificity (ability to correctly identify negative cases), we prioritize sensitivity as losing recurring revenue and the lifetime value of a customer costs the company more than giving out an unnecessary retention incentive.

The predictive models selected for this project were chosen with the intention of representing a wide range of classification techniques. All of the models have varying degrees of complexity and interpretability. Decision trees are highly intuitive and easy for stakeholders to understand. Random forests improve predictive performance by combining multiple decision trees. Neural networks and support vector machines were also explored because they can capture more complex, non-linear relationships in the data. Using several model types allows our team to compare performance while keeping a balance of predictive accuracy, sensitivity to noise, and interpretability to identify the best solution for Telcom Inc.

4.2 Model Development

Developed using the same strategy as the descriptive tree, the predictive decision tree was grown to its maximum size on the training data then pruned using the minimum and standard error summation technique. This process yielded a tree with a complexity parameter of 0.005 and 8 split points (Figure 4.2.1). Classification thresholds were tuned on this pruned tree, resulting in an optimal threshold of 0.25.

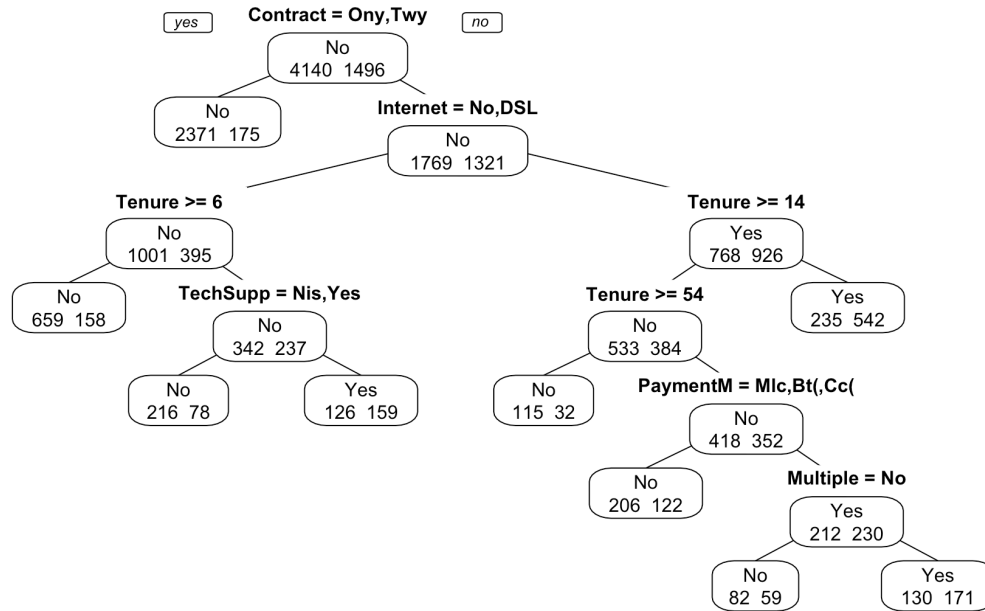


Figure 4.2.1. Pruned predictive decision tree.

To improve the predictive power of a single decision tree, we implemented a random forest ensemble model. This algorithm uses multiple independent decision trees that are formed on random subsets of the training data and averages their outcomes. This model was trained on different minimum node sizes to prevent overfitting, resulting in an optimal value of 10. This means that each leaf node (or end point where records receive a classification) must include at least ten records. Classification thresholds were tuned on the ensemble, resulting in an optimal threshold of 0.25.

The neural network is one of the models that requires structured, numeric data in order to function properly. This model was trained and tested on data transformed via the process detailed in Section 2.1. For this model, we used a logistic activation function as it is ideal for binary classification. Different hidden layer and node counts were tested before optimizing at a single

layer with six hidden nodes (Figure 4.2.2). Raw probabilities were extracted from the output and evaluated against different thresholds, establishing an ideal cutoff of 0.25.

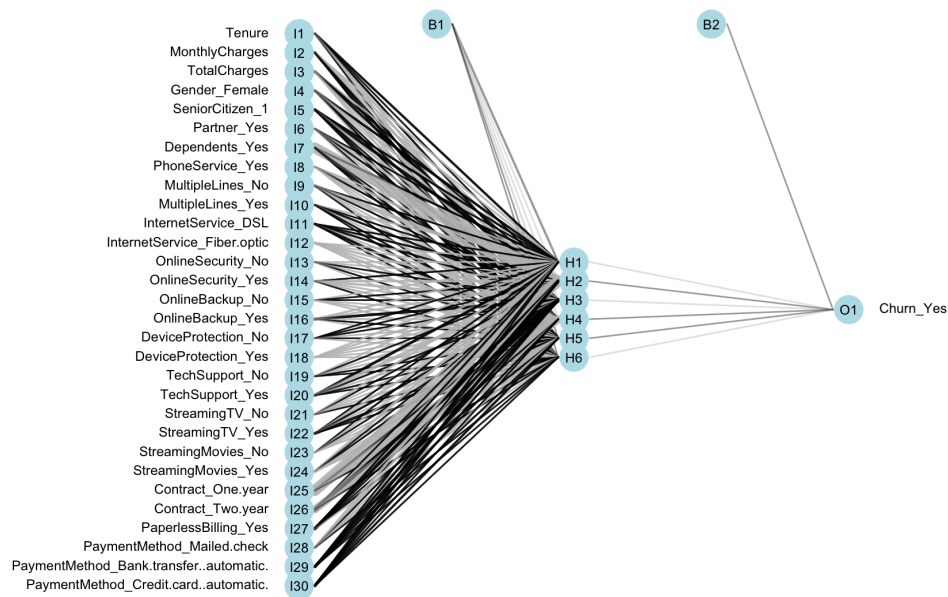


Figure 4.2.2. Neural network structure.

Our final model, the support vector machine (SVM), was also trained and tested on the transformed data. The SVM uses the standard radial basis kernel, as it can handle non-linear data. A two-stage tuning process was executed in order to find the best values for the gamma and cost hyperparameters. The initial search tested a wider range of gamma and cost values, with the second round focusing on values closer to the first set of optimal values. This optimization resulted in a final model with a gamma of 0.05 and a cost of 0.5. Class probabilities were then generated and evaluated against different thresholds, establishing a cutoff at 0.25.

4.3 Model Performance

Despite being separately trained, each model optimized at the same probability threshold of 0.25. This lower cutoff threshold delivers a solid balance between overall accuracy and sensitivity (Table 4.3.1). Coincidentally, this also makes direct comparison of model performance easier.

Model	Accuracy	Sensitivity
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Decision Tree	75.84%	79.09%
Random Forest	75.20%	75.87%
Neural Network	74.13%	69.17%
SVM	77.83%	62.73%

Table 4.3.1. Predictive model performance at a probability threshold of 0.25.

Although all of the models achieved similar accuracy and sensitivity scores, the decision tree has the most well-rounded performance. Although its overall accuracy is marginally lower than that of the SVM, its sensitivity is the highest of all the models, which is our priority. The decision tree's balance of accuracy and sensitivity makes it the best choice for the predictive model. Therefore, the predictive model deliverable for this project is a decision tree. It is worth noting that from a technical standpoint, you may think that the random forest model should outperform a single decision tree, given it's an ensemble. But given the simplicity of the data (not too many features and only two class levels), the single decision tree might just suit the data better in the same way that the logistic regression model does not. In the future, if the available data becomes more robust, the models will be re-evaluated to see if there is a better fit.

5. Insights and Recommendations

Based on the retention and churn drivers identified by the descriptive models, we can offer some data-backed strategies to improve retention and proactively prevent churn. First, Telcom should look to prioritize selling contracts opposed to month-to-month service, specifically one-to-two-year contracts. One way to implement this is to encourage customers with month-to-month plans to switch to long-term contracts by offering discounts or a first month free promotion, as well as steering new customers to contracted plans. Additionally, a shift to automated billing, combined with incentives for using automated billing (ex. discounts), can also help improve customer retention. With customers who use tech support also having lower churn tendencies, this indicates that fostering relationships with customers should be a priority for all client-facing employees. The marketing and customer service departments should be made aware of these findings to allow for strategic campaign and promotion planning.

Product wise, Telcom should work to improve internet services. As a service that supports many other Telcom products (ex. security, backup, media streaming), it is crucial to provide a high-quality experience to customers. Losing a customer who uses Telcom's internet

services also means a loss of revenue from these other areas. The IT and network specialists should prioritize improving the quality of internet services, while also ensuring to maintain phone line quality.

Through the datasets and models, we discovered that demographic factors are not significant in predicting churn, and that customer experience is the main deciding factor. Therefore, promotional efforts should not necessarily target customers by demographic factors, but rather by behavioral and usage factors. However, these factors can still be considered when dealing with the specifics of promotions. For example, the contract type may be the determining factor in *which* customers are targeted for promotions to switch to long-term contracts, but demographic information like age can be used to determine *what* that promotion is. These promotions would be time sensitive, depending on what is currently considered “on trend” for different demographics. For more detailed recommendations and specific plans, see the Retention Playbook.

6. Model Deployment

Deploying the predictive model involves making it accessible to marketing as well as customer support and sales representatives. For marketing, a script within the database of customer information will run on a schedule (ex. weekly). Customer data will run through the predictive model and classify each customer as likely to churn or not. Customers crossing the current threshold of 25% will be tagged, triggering their enrollment in retention specific marketing (see Retention Playbook).

For customer support and sales representatives, these tags will automatically sync into Telcom’s CRM software. When a representative accesses a customer’s profile, a flag will display a specific recommendation tactic, such as discounts and contract incentives (see Retention Playbook). The Retention Playbook can also be referenced for additional strategies while dealing with customers directly.

This deployment is supported by the Retention Playbook, shared across the customer service, marketing, and analytics teams. Using information from the descriptive models, the playbook maps churn drivers to approved promotions and scripts specifically designed to combat that driver. The analytics team will monitor the models to ensure that the right factors are being

addressed in the retention tactics being developed, while marketing and customer service leadership will review these tactics as the market shifts.

6.1 Lifecycle Management & Business Benefit Over Time

Lifecycle management of the model will be important to ensure it continues to provide accurate reports and forecasting on customer churn over time. It will be important to update the model with new data or findings regularly. As Telcom evolves as a company, there may be changes to products and services offered, potentially resulting in different data set attributes. As the data changes, the models should be re-evaluated and re-tuned to determine if there is a better fit (ex. neural network, SVM).

To measure the success of the model and retention strategies suggested, we must find metrics to determine whether the model is helpful in identifying why customers are likely to churn and if our recommended strategies based on this model are actually useful in limiting churn. We will measure the financial ROI and operational impact of the new retention strategies. Success will be measured using metrics including overall churn rate reduction, retained recurring revenue (i.e. from preventing a churn), and the net return on retention campaigns (saved lifetime value versus cost of discounts). Comparing high-risk customers who receive proactive intervention and an uncontacted group through A/B testing will also help Telcom determine the project's direct contribution to retention. Exit surveys will also provide insight as to why customers are leaving, and this could be compared to our model indicators. Over time, this information can be incorporated into our models to add another layer behind our predictions and directly influence retention strategies to ensure tactics are aligned with customer needs.

Appendix A

Table A1

Description of variables in the project dataset.

Variable	Description
customerID	Unique customer ID
Gender	Indicates male or female
Senior Citizen	Indicates if customer is senior citizen (1) or not (0)
Partner	Indicates if the customer has a marital partner (Yes, No)
Dependents	Indicates if the customer has dependents (Yes, No)
Tenure	Number of months the customer has been with Telcom Inc.
PhoneService	Indicates if the customer has a phone service (Yes, No)
MultipleLines	Indicates if the customer has multiple phone lines (Yes, No, No phone service)
InternetService	Customer's type of internet service (DSL, Fiber optic, No)
OnlineSecurity	Indicates if the customer has online security (Yes, No, No internet service)
OnlineBackup	Indicates if the customer has online backup (Yes, No, No internet service)
DeviceProtection	Indicates if the customer has device protection (Yes, No, No internet service)
TechSupport	Indicates if the customer has tech support (Yes, No, No internet service)
StreamingTV	Indicates if the customer has streaming TV (Yes, No, No internet service)
StreamingMovies	Indicates if the customer has streaming movies (Yes, No, No internet service)
Contract	The contract term of the customer (Month-to-month, One year, Two year)
PaperlessBilling	Indicates if the customer has paperless billing (Yes, No)
PaymentMethod	The customer's payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic))
MonthlyCharges	The amount charged in dollars to the customer monthly
TotalCharges	The total cumulative amount charged in dollars to the customer
Churn	Indicates if the customer churned (Yes, No)

Appendix B

Model Code

A separate .R file containing all code is available to run.

```
library(dplyr)
library(readr)
library(caret)
library(rpart)
library(rpart.plot)

# read in the data, specify appropriate factor levels
telco <- read_csv("telco.csv", col_types = cols(
  gender = col_factor(levels = c("Male", "Female")),
  SeniorCitizen = col_factor(levels = c("0", "1")),
  Partner = col_factor(levels = c("No", "Yes")),
  Dependents = col_factor(levels = c("No", "Yes")),
  PhoneService = col_factor(levels = c("No", "Yes")),
  MultipleLines = col_factor(levels = c("No phone service", "No", "Yes")),
  InternetService = col_factor(levels = c("No", "DSL", "Fiber optic")),
  OnlineSecurity = col_factor(levels = c("No internet service", "No", "Yes")),
  OnlineBackup = col_factor(levels = c("No internet service", "No", "Yes")),
  DeviceProtection = col_factor(levels = c("No internet service", "No", "Yes")),
  TechSupport = col_factor(levels = c("No internet service", "No", "Yes")),
  StreamingTV = col_factor(levels = c("No internet service", "No", "Yes")),
  StreamingMovies = col_factor(levels = c("No internet service", "No", "Yes")),
  Contract = col_factor(levels = c("Month-to-month", "One year", "Two year")),
  PaperlessBilling = col_factor(levels = c("No", "Yes")),
  PaymentMethod = col_factor(levels = c("Electronic check", "Mailed check", "Bank
transfer (automatic)", "Credit card (automatic)")),
  Churn = col_factor(levels = c("No", "Yes"))))

# remove customer ID column and make capitalization consistent
telco <- telco %>% select(-1) %>%
  rename(Gender = gender,
         Tenure = tenure)

# replace missing total charges with 0s (have not been billed yet)
telco$TotalCharges[is.na(telco$TotalCharges)] = 0

# descriptive model: logistic regression
full_model = glm(Churn~., data = telco, family = "binomial")
summary(full_model)

mod_backward = step(full_model, direction = "backward", k = 2, trace = 0)
summary(mod_backward)

# Tenure: higher = decreases
# Internet service: having = increases
# One and two year contracts = decreases (vs month to month)
# Paperless billing: having = increases
# Payment method other than electronic check = decreases
# Total charges: higher = increases

# descriptive model: decision tree
set.seed(19)
fulltree = rpart(Churn~., data = telco, method = "class", cp = 0, minsplit = 2,
minbucket = 1, maxsurrogate = 0)
prp(fulltree, type = 1, extra = 1)
printcp(fulltree)

# find the best pruned tree:
# min error tree = 0.77421 with xstd = 0.018142
```

```

# min xerror + xstd = 0.792352
bestpruned = prune(fulltree, cp = 0.007)
prp(bestpruned, type = 1, extra = 1)
printcp(bestpruned)
# contract: one year or two year = less churn
# internet service: none or DSL = less churn (vs fiber optic)
# tenure: longer = less churn
# tech support: using tech support = less churn (encourage people to reach out if they
have problems)
# payment method: not a crucial factor but electronic check tends to lead to higher
churn path
# multiple lines: more of an end factor too but multiple lines = churn

# split training and testing data
set.seed(19)
train_index = createDataPartition(telco$Churn, p = 0.8, list = FALSE)

train = telco[train_index,]
test = telco[-train_index,]

# predictive model: logistic regression
fullmodel = glm(Churn~., data = train, family = "binomial")
summary(fullmodel)

stepmodel = step(fullmodel, direction = "backward", k = 2, trace = 0)
summary(stepmodel)

# tune threshold then test
logmodel_prob = predict(stepmodel, type = "response")
logmodel_class = ifelse(logmodel_prob >= 0.4, 1, 2)
logmodel_class = factor(logmodel_class, labels = c("No", "Yes"))
logmodel_matrix = confusionMatrix(logmodel_class, train$Churn, positive = "Yes")
logmodel_matrix

# test
logmodel_prob = predict(stepmodel, type = "response", newdata = test)
logmodel_class = ifelse(logmodel_prob >= 0.4, 1, 2)
logmodel_class = factor(logmodel_class, labels = c("No", "Yes"))
logmodel_matrix = confusionMatrix(logmodel_class, test$Churn, positive = "Yes")
logmodel_matrix

# predictive model: decision tree
set.seed(19)
fulltree = rpart(Churn~., data = train, method = "class", cp = 0, minsplit = 2,
minbucket = 1, maxsurrogate = 0)
prp(fulltree, type = 1, extra = 1)
printcp(fulltree)

# find the best pruned tree:
# min error tree = 0.78008 with xstd = 0.020334
# min xerror + xstd = 0.800414

bestpruned = prune(fulltree, cp = 0.00502)
prp(bestpruned, type = 1, extra = 1)
printcp(bestpruned)

# tune threshold then test
tree_prob = predict(bestpruned, type = "prob")[,2]
tree_class = ifelse(tree_prob >= 0.25, 2, 1)
tree_class = factor(tree_class, labels = c("No", "Yes"))
tree_matrix = confusionMatrix(tree_class, train$Churn, positive = "Yes")
tree_matrix

```

```

# test
tree_prob = predict(bestpruned, newdata = test, type = "prob")[,2]
tree_class = ifelse(tree_prob >= 0.25, 2, 1)
tree_class = factor(tree_class, labels = c("No", "Yes"))
tree_matrix = confusionMatrix(tree_class, test$Churn, positive = "Yes")
tree_matrix

# predictive model: random forest
library(randomForest)
set.seed(19)

# trained with different node sizes
rf_model <- randomForest(Churn ~ ., data = train, importance = TRUE, nodesize = 10)

# predict probabilities
rf_train_prob = predict(rf_model, train, type = "prob")[, "Yes"]

# convert raw probabilities to a 0 or 1 class using threshold (test)
rf_train_class = ifelse(rf_train_prob >= 0.25, "Yes", "No")
rf_train_class = factor(rf_train_class, levels = c("No", "Yes"))

# confusion matrix for training data
confusionMatrix(rf_train_class, train$Churn, positive = "Yes")

# predict probabilities on test data
rf_test_prob = predict(rf_model, test, type = "prob")[, "Yes"]

# apply threshold
rf_test_class = ifelse(rf_test_prob >= 0.25, "Yes", "No")
rf_test_class = factor(rf_test_class, levels = c("No", "Yes"))

# generate confusion matrix for testing data
confusionMatrix(rf_test_class, test$Churn, positive = "Yes")

# predictive model: neural network
library(neuralnet)
library(fastDummies)

# transform the data
# convert characters/factors to dummy columns automatically
telco_nn <- dummy_cols(train, remove_selected_columns = TRUE, remove_first_dummy = TRUE)
test_nn <- dummy_cols(test, remove_selected_columns = TRUE, remove_first_dummy = TRUE)

# rename columns to ensure there are no spaces or special characters for formula compatibility
names(telco_nn) <- make.names(names(telco_nn))
names(test_nn) <- make.names(names(test_nn))

# scale numeric columns
num_cols <- c("Tenure", "MonthlyCharges", "TotalCharges")
telco_nn[num_cols] <- scale(telco_nn[num_cols])
test_nn[num_cols] <- scale(test_nn[num_cols])

# train with different hidden layers/nodes
set.seed(19)
nn_model <- neuralnet(
  Churn_Yes ~ .,
  data = telco_nn,
  act.fct = "logistic",
  hidden = 6,
  linear.output = FALSE,

```

```

    stepmax = 1e6
)

# plot of network
library(NeuralNetTools)
dev.new(width = 12, height = 8)
plotnet(
  nn_model,
  circle_cex = 2.5,
  cex_val = 0.55,
  pad_x = 0.6,
  pos_col = "black",
  neg_col = "grey70",
  alpha_val = 0.4
)

# predict probabilities on validation data
valid.pred = predict(nn_model, select(test_nn, ~"Churn_Yes"))

# convert raw probabilities to a 0 or 1 class using threshold
valid.class = ifelse(valid.pred > 0.25, 1, 0)

# generate the confusion matrix using actual validation labels
table(valid.class, test_nn$Churn_Yes)

# calculate overall accuracy
mean(valid.class == test_nn$Churn_Yes)

# predictive model: SVM
library(e1071)
set.seed(19)

# find best parameter values for cost and gamma (using transformed data)
tune.out=tune(svm,as.factor(Churn_Yes)~.,data = telco_nn,kernel="radial",
              ranges=list(cost=c(0.001,0.01,0.1,1,10,100),
                          gamma = c(0.001, 0.01, 0.1, 1)))
summary(tune.out)

bestmod = tune.out$best.model
summary(bestmod) # gamma = 0.01, cost = 1

# further tune around those values
set.seed(19)
tune.out_tuned=tune(svm,as.factor(Churn_Yes)~.,data = telco_nn,kernel="radial",
                   ranges=list(cost=c(0.25, 0.5, 0.75, 1, 1.25, 1.5, 2),
                               gamma = c(0.05, 0.08, 0.1, 0.2, 0.3)))

bestmod = tune.out_tuned$best.model
summary(bestmod)
# gamma = 0.05, cost = 0.5

set.seed(19)
svmfit.opt=svm(as.factor(Churn_Yes)~.,data=telco_nn, kernel="radial",
               gamma=bestmod$gamma, cost=bestmod$cost, probability=TRUE)

# predict probabilities
ypred=predict(svmfit.opt,newdata=select(test_nn, ~"Churn_Yes"), probability = TRUE)
yprob = attributes(ypred)$probabilities
churn_probabilities = yprob[, "1"]

# use threshold to get class
ypred_high_sensitivity = ifelse(churn_probabilities > 0.25, 1, 0)

```

```
# generate confusion matrix
table(pred = ypred_high_sensitivity, true = test_nn$Churn_Yes)
mean(ypred_high_sensitivity == test_nn$Churn_Yes)

TP <- sum(ypred_high_sensitivity == 1 & test_nn$Churn_Yes == 1)
FN <- sum(ypred_high_sensitivity == 0 & test_nn$Churn_Yes == 1)
TP / (TP + FN)
```

Works Cited

BlastChar. (2017). Telco Customer Churn. Kaggle.

<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Kumar, S. (2022, December 12). Customer Retention versus Customer Acquisition. *Forbes*.

<https://www.forbes.com/councils/forbesbusinesscouncil/2022/12/12/customer-retention-versus-customer-acquisition/>